MIL-R-25536A(USAF)
6 April 1959
Superseding
MIL-E-25536(USAF)
14 March 1956

MILITARY SPECIFICATION

ROCKET MOTORS, AERONAUTICAL, ACCEPTANCE TEST FOR

1. SCOPE

1.1 This specification establishes the acceptance test requirements for rocket motors for aeronautical vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-R-25532	Rocket Motors, Aeronautical, General Specification For
MIL-R-25534	Rocket Motors, Aeronautical, Qualification Test For

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables For Inspection By Attributes
MIL-STD-414	Sampling Procedures and Tables For Inspection By Variables For Percent Defective

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 ACCEPTANCE TEST DATA, ROCKET MOTOR AND COMPONENTS.- Rocket motor and component acceptance test data shall be prepared as follows:

FSC 1336
FSC 1340

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3.1.1 GENERAL

3.1.1.1 DIMENSIONAL UNITS.- All dimensions shall be expressed in the English gravitational system of units.

3.1.1.2 TEST DATA CORRECTIONS

3.1.1.2.1 TEMPERATURE.- Performance characteristics shall not be corrected for free-air temperature.

3.1.1.2.2 PRESSURE.- An approximate correction for barometric pressure altitude shall be applied to thrust and impulse values. This correction shall be made as follows:

$$F_{sl}=F_{alt}- A_e(P_{sl}- P_{alt}) \qquad I_{sl}=I_{alt}x \frac{F_{sl}}{F_{alt}}$$

3.1.1.3 ACCEPTANCE DATA SUMMARY SHEET.- An acceptance data summary sheet shall be prepared in accordance with figure 1 for each rocket motor. Sections I, II, and the applicable items of sections III and IX shall be completed for all rocket motors. The remainder of the sheet shall be completed for any rocket motors fired. The calibration data for any calibrated components shall be attached to this sheet.

3.2 DISPOSITION OF ACCEPTANCE DATA.- Acceptance test data and acceptance summary data sheets shall be retained by the testing agency for 5 years and furnished to the procuring activity or authorized representative upon request.

4. QUALITY ASSURANCE PROVISIONS

4.1 GENERAL.- Complete specifications for the rocket motor and all components shall be furnished to the procuring activity prior to beginning the acceptance tests. Copies of all contractor instructions, procedures, specifications, and drawings governing the inspection and testing of the rocket motor and its components shall be kept on file by the contractor and shall be supplied to the procuring activity upon request. When instructions, procedures, specifications, or drawings have been supplied to the procuring activity all changes shall also be supplied upon release. At convenient times prior to the tests and after the tests the rocket motors and components shall be examined to determine if they conform to all requirements of the contract and specifications under which they were built.

4.1.1 TEST METHOD AND APPARATUS.- Acceptance test method and apparatus used shall be in accordance with that specified in the approved test method document required by MIL-R-25532.

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FIGURE 1

Data Summary Sheet

Test conducted by (name of contractor) Date _____

Contract No. _____ Purpose _____

I ROCKET MOTOR INFORMATION

Rocket Motor Model Designation _____
Rocket Motor Lot Number _____ Rocket Motor Sublot Number _____
Rocket Motor Serial No. _____ Case Serial No. _____
Nozzle Serial No. _____
Propellant Type or Designation _____
Propellant Lot No. _____ Propellant Batch No(s) _____
Propellent Grain Serial No. _____

II IGNITER INFORMATION

Type _____
Lot No. _____ Sublot No. _____ Igniter Serial No. _____
Squib resistance before conditioning _____ after conditioning _____

III PHYSICAL DATA

Rocket Motor Weight - before firing _____
- after firing _____
Nozzle Expansion Ratio _____
Nozzle Throat Area - before firing _____
- after firing _____
Center of Gravity Location - before firing _____
- after firing _____

IV PRETEST TREATMENT

Conditioned Temperature _____

V ACTUAL TEST CONDITIONS

Time removed from conditioning _____ time fired _____
Observed Barometric Pressure _____
Temperature of Barometer _____
Corrected Barometric Pressure _____
Ambient Air Temperature at firing _____

VI POST FIRING INSPECTION RESULTS

(Include results of post firing inspection)

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FIGURE 1
(continued)

VII REDUCED AND DERIVED DATA

Action time total impulse _____
Burn time total impulse _____
Deliverable total impulse _____
Specific impulse _____
Overall specific impulse _____
Maximum thrust _____
Maximum ignition thrust _____
Average thrust _____
Maximum chamber pressure _____
Maximum ignition pressure _____
Average chamber pressure _____
Action time _____
Burn time _____
Ignition delay time _____
Ignition rise time _____
Burn rate _____
Ignition current used _____

VIII PERFORMANCE TRACES

Thrust-time trace with point of firing switch actuation shown
Pressure-time trace with point of firing switch actuation shown
Hi-speed ignition thrust-time trace with point of firing switch actuation shown
Hi-speed ignition pressure-time trace with point of firing switch actuation shown

IX THRUST CONTROL DEVICES

Device Model Designation
Device Lot Number _____ Device Sublot Number _____
Device Serial Number _____
Squib Resistence _____

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4.1.1.1 ACCURACY OF DATA.- For all rocket motor and component tests, the test apparatus shall be such as to insure that reported data will have a steady state accuracy of ± 2 percent of the maximum rated value, except that temperature shall be accurate to ± 2°F and weight shall be accurate to ± 0.2 percent. All apparatus shall be calibrated often enough to insure that this degree of accuracy is maintained. Calibration records shall be retained by the testing agency for 2 years, and furnished to the procuring activity upon request.

4.1.1.2 AUTOMATIC RECORDING EQUIPMENT.- Automatic recording equipment of adequate response shall be used to obtain data during those parts of the rocket motor and component tests requiring the evaluation of time versus rocket motor or component variables.

4.1.2 TEST CONDITIONS.- The following test conditions shall apply unless otherwise specified herein.

4.1.2.1 TEMPERATURES.- Unless otherwise specified all tests shall be conducted at room temperature as defined in 4.1.2.1.1.

4.1.2.1.1 ROOM TEMPERATURE.- Room temperature tests shall be conducted with the temperature of the rocket motor or component within the range of 10°F to 110°F.

4.1.2.1.2 LOW TEMPERATURE.- Low temperature tests shall be conducted with the temperature of the rocket motor or component at -75°F or lower.

4.1.2.1.3 NORMAL TEMPERATURE.- Normal temperature tests shall be conducted with the temperature of the rocket motor or component at 59°F ± 5°F.

4.1.2.1.4 HIGH TEMPERATURE.- High temperature tests shall be conducted with the temperature of the rocket motor or component at +170°F or higher.

4.1.2.2 PRESSURE.- Unless otherwise specified all tests shall be conducted at ambient atmospheric pressure.

4.1.2.2.1 CHAMBER PRESSURE.- Chamber pressure measurements shall be made at the head end of the rocket motor.

4.1.2.3 TIMES

4.1.2.3.1 CONDITIONING TIME.- Conditioning time for the rocket motor shall be such that all parts of the rocket motor shall have reached a temperature within 5°F of the specified temperature. A rocket

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motor shall be considered conditioned when it has been continuously exposed to the specified temperature for the conditioning time, making suitable allowance for the starting temperature. During the conditioning time the temperature of the conditioning chamber shall not vary more than ± 5°F from the specified temperature.

4.1.2.3.2 TRANSFER TIME.- The rocket motor transfer time shall not exceed that specified in the model specification.

4.2 ACCEPTANCE TESTS.- Each lot of rocket motors submitted for acceptance shall be subjected to the acceptance tests specified herein.

4.2.1 LOT.- A lot of rocket motors shall consist of a number of rocket motors produced in a continuous production period with no change in process or in raw material lots. The inspection lot for non-destructive inspection within the above limitation shall be designated by the procuring activity and may differ from the quantity designated in the contract or order as a lot for production, shipment, or other purpose.

4.2.2 ROCKET MOTOR COMPONENTS

4.2.2.1 SAMPLING AND INSPECTION.- Where applicable, sampling plans and methods conforming to the provisions of MIL-STD-105 shall be used for components and individual parts subjected to nondestructive inspection. The contractor shall submit these plans and methods to the procuring activity for approval as part of the detailed acceptance test method. As a part of these plans and methods the contractor shall prepare a "Classification of Defects" for each component and part thereof in accordance with the definitions of classes of quality characteristics given in MIL-STD-105.

4.2.2.1.1 PROPELLANT.- The contractor shall sample and perform ballistic and physical property tests on each batch of propellant to determine if the batch meets the requirements of the specification under which it was produced.

4.2.2.1.1.1 BATCH SIZE.- Batch size shall be designated by the contractor and approved by the procuring activity.

4.2.2.1.2 GRAIN.- Each finished grain shall be subjected to radiographic, ultrasonic, or fluoroscopic inspection in addition to a visual inspection. Any voids, cracks or low density areas observed in the grain that would materially affect the ballistic performance of the rocket motor or cause an unsafe condition upon firing shall be cause for rejection.

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4.2.2.1.3 CASE.- All rocket motor cases shall be subjected to a static proof pressure test. Any evidence of more than 0.2 percent permanent deformation, distortion, leakage, cracking, or other failure during this test shall be cause for rejection. In addition all welds shall be radiographically inspected.

4.2.2.1.4 SAFETY DIAPHRAGM ASSEMBLY.- All safety diaphragm assemblies shall be subjected to a static proof pressure test. Any evidence of more than 0.2% permanent deformation, distortion, leakage, cracking, or other failure during this test shall be cause for rejection.

4.2.2.1.5 ATTACHMENT FITTINGS.- All rocket motor and thrust control device attachment fittings shall be subjected to magnetic particle, fluorescent penetrant, or radiographic inspection. Any evidence of cracks shall be cause for rejection.

4.2.2.1.6 IGNITERS, SQUIBS AND EXPLOSIVE COMPONENTS.- All igniters, squibs, and explosive components shall be tested for proper electrical resistance.

4.2.2.1.7 THRUST CONTROL DEVICES.- All thrust control devices shall be subjected to suitable tests to insure proper operation. Adequate provisions shall be made to insure that calibrated components are within calibration limits.

4.2.2.1.8 ELECTRICAL, HYDRAULIC, AND PNEUMATIC SYSTEMS.- All electrical, hydraulic, and pneumatic system components shall be subjected to suitable tests to insure proper operation. Adequate provisions shall be made to insure that calibrated components are within calibration limits.

4.2.2.1.9 STANDARD PARTS.- Commercial, AN, and MS standard parts such as cotter pins, washers, and similar low-stressed items require only a visual inspection.

4.2.3 ROCKET MOTORS

4.2.3.1 POST ASSEMBLY INSPECTIONS

4.2.3.1.1 ATTACHMENT-FITTINGS.- Attachment fittings of all rocket motors shall be inspected to assure that installation dimensions are within drawing limits.

4.2.3.1.2 NOZZLE ALIGNMENT.- The nozzle alignment of all rocket motors shall be inspected to assure that it is within drawing limits.

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4.2.3.1.3 CENTER OF GRAVITY.- The center of gravity location of all rocket motors shall be determined to assure that it is within drawing limits.

4.2.3.1.4 REJECTION AND RETEST.- Rocket motors rejected during the inspections in 4.2.3.1.1, 4.2.3.1.2, or 4.2.3.1.3 may be resubmitted for acceptance after rework. Before resubmission full particulars concerning previous rejection(s), and corrective action taken, shall be furnished the Inspector.

4.2.3.2 FIRING TEST.- Firing tests shall be conducted on rocket motors submitted for acceptance test.

4.2.3.2.1 SAMPLING.- The acceptance test sampling method for firing tests (destructive testing) shall be designed by the contractor and submitted to the procuring activity for approval as part of the detailed acceptance test method. The acceptance test sampling methods proposed by the contractor shall be described in detail, including the sampling risks involved (operating characteristic curves). The variables sampling methods of MIL-STD-414 shall be used whenever possible.

4.2.3.2.2 FIRING TEMPERATURE SCHEDULE.- Firing tests shall be conducted in accordance with the following schedule:

a. A random selection shall be made of the static firing temperature (high, normal or low) for sample rocket motors from the first lot.

b. A random selection shall be made of the static firing temperature from the two remaining temperatures for the sample rocket motors from the second lot.

c. The sample rocket motors from the third lot shall be static fired at the remaining temperature.

The above schedule shall be repeated for the succeeding lots.

4.2.3.2.3 POST-FIRING INSPECTION.- After firing each rocket motor shall be inspected for evidence of mechanical malfunction or failure.

4.2.4 ADDITIONAL TESTS.- Additional rocket motor or component tests for the purpose of acceptance testing special features of the rocket motor may be required by the procuring activity. These tests shall be specified in the model specification.

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4.2.5 PREPRODUCTION TEST.- The first 27 rocket motors from a production run may be subjected to a preproduction test at the discretion of the procuring activity. The purpose of this test is to provide assurance that the rocket motors produced in quantity are of the same characteristics as those submitted for qualification test. The test schedule shall consist of those tests specified for test groups I through VI inclusive of MIL-R-25534. The test method shall be the same method approved for use in the qualification test.

4.2.6 ACCEPTANCE CONDITIONS.- Acceptance of a lot of rocket motors shall be predicated upon meeting all requirements of the acceptance plan.

5. NOTES

5.1 SYMBOLS AND DEFINITIONS.- The symbols and definitions used in this specification are as specified in MIL-R-25532.

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